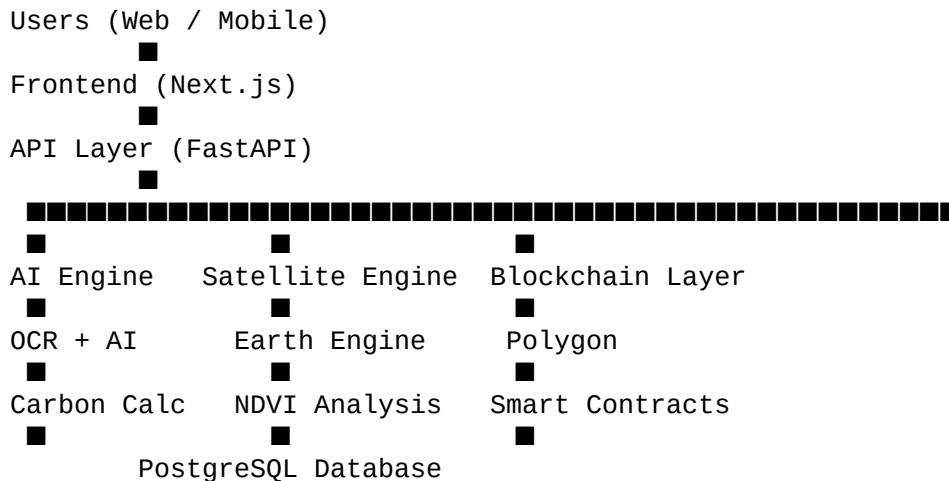


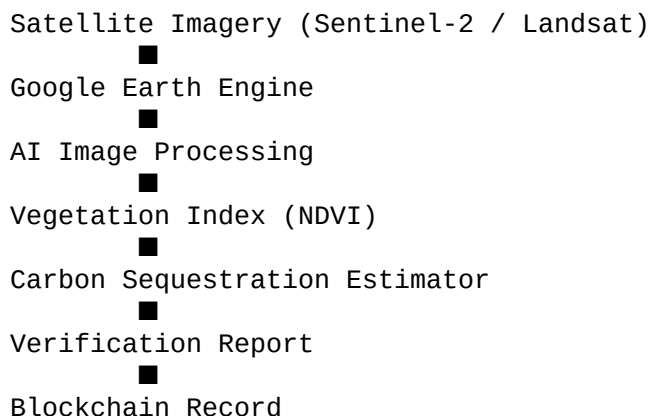
Blockchain Carbon Credit Trading Platform – Architecture Diagrams

1. High-Level System Architecture



- Frontend shows dashboards, carbon marketplace, and scanning tools
- Backend API handles logic and requests
- AI Engine processes scanned documents and calculates emissions
- Satellite Engine verifies environmental projects
- Blockchain layer manages tokenized carbon credits

2. Satellite Verification Pipeline



- Uses satellite data to verify environmental projects
- NDVI index measures vegetation health
- Carbon absorption is estimated from vegetation density
- Verification results are stored on blockchain

3. AI Carbon Footprint Scanner Pipeline

User Uploads Image
(Receipt / Bill / Ticket)



OCR Engine
(Tesseract / Vision API)



Text Extraction



AI Classification



Carbon Calculation Engine



Emission Report



Carbon Credit Recommendation

- OCR extracts text from scanned documents
- AI detects activity type such as travel or electricity usage
- Carbon calculator converts activity to CO2 emissions
- System recommends carbon credits to offset emissions

4. Blockchain Carbon Credit Flow

Environmental Project



Verification (Satellite + AI)



Carbon Credit Issuance



Smart Contract (Polygon)



Tokenized Carbon Credit



Marketplace Trading



Credit Retirement

- Projects are verified before credits are issued
- Smart contracts mint carbon credit tokens
- Users can buy or sell credits in the marketplace
- Credits are retired when used to offset emissions